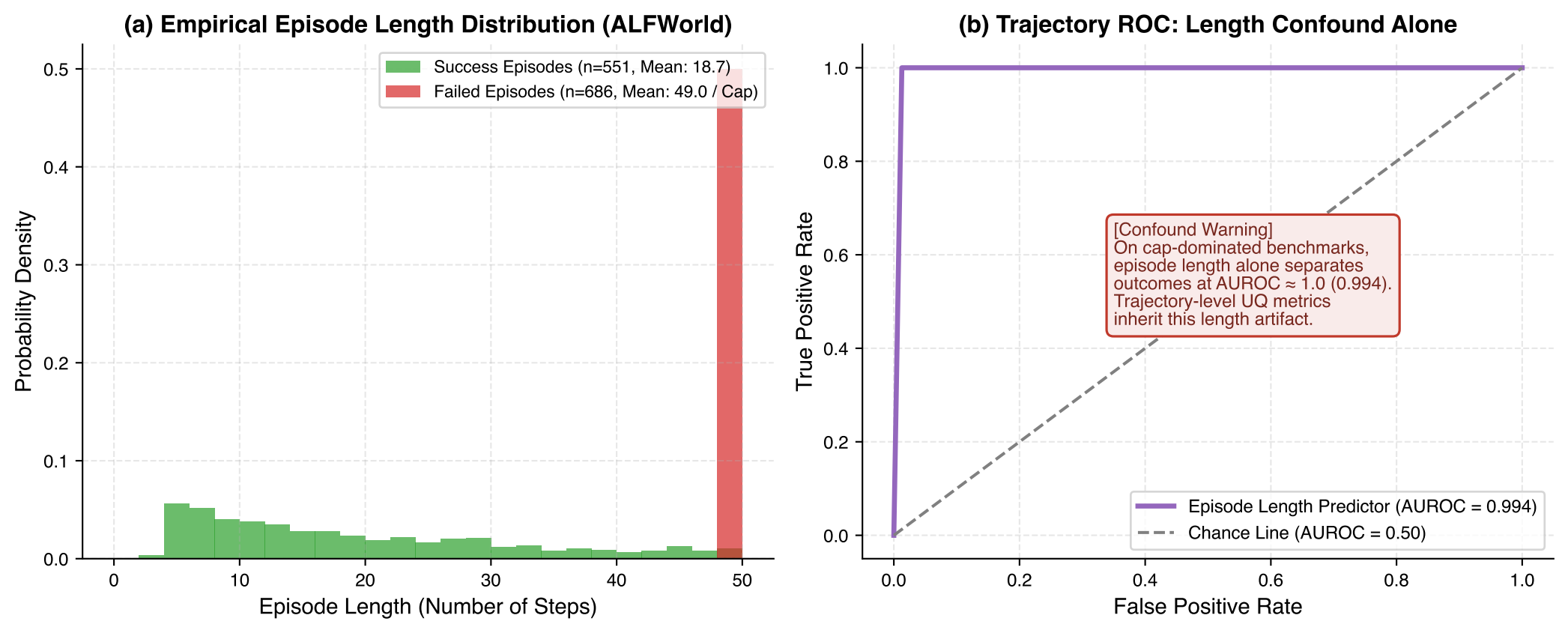
(a) Empirical Episode Length Distribution (ALFWorld)
Success Episodes (n=551, Mean: 18.7)
Failed Episodes (n=686, Mean: 49.0 / Cap)
Probability Density
Episode Length (Number of Steps)
0.0
0.1
0.2
0.3
0.4
0.5
0
10
20
30
40
50
(b) Trajectory ROC: Length Confound Alone
[Confound Warning]
On cap-dominated benchmarks,
episode length alone separates
outcomes at AUROC ≈ 1.0 (0.994).
Trajectory-level UQ metrics
inherit this length artifact.
Episode Length Predictor (AUROC = 0.994)
Chance Line (AUROC = 0.50)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0